

ODD 2024 Tutorial Sheet - Functions

Software Development Fundamentals – I (15B11CI111)

Course Outcomes (CO)	
CO1	Explain various phases of software development life cycle
CO2	Explain various data types, memory allocation schemes, precedence of arithmetical and logical operations, and need of array, and structures
CO3	Draw the flow chart and write the high level code for different problems
CO4	Apply and implement functions with or without pointers for different problems
CO5	Demonstrate and implement various operations like traverse, insertion, deletion, etc. on files

Note: Students are advised to submit their solutions to their respective tutorial faculties

Q1. [CO4] Answer the following (a) and (b) both.

- (a) What are actual and formal parameters in call by value functions?
- (b) What are the features of call by value functions?

Q2. [CO4] Write a program in C to find the sum of the series $1! / 1+2! / 2+3! / 3+4! / 4+5! / 5$ using the function.

Q3. [CO4] What will be the output of the following program?

```
#include<stdio.h>
void modify (int n) {
n=n+10;
printf("Value of 'n' inside function=%d \n", n);
}

int main() {
int x=15;
modify(x);//passing value in function
printf("Modified value of x=%d \n", x);
return 0;
}
```

Q4. [CO4] A girl carries 50 hair pins in a haphazard manner within her dressing drawer. Her mother asked her to arrange these pins so that the drawer will look elegant. But the girl is confused as she does not know the number of possible ways in which she can arrange her hair pins. You must suggest a recursive function to the girl in order to ease her task.

Q5. [CO4] Write a program in C to convert decimal number to binary number using the function.